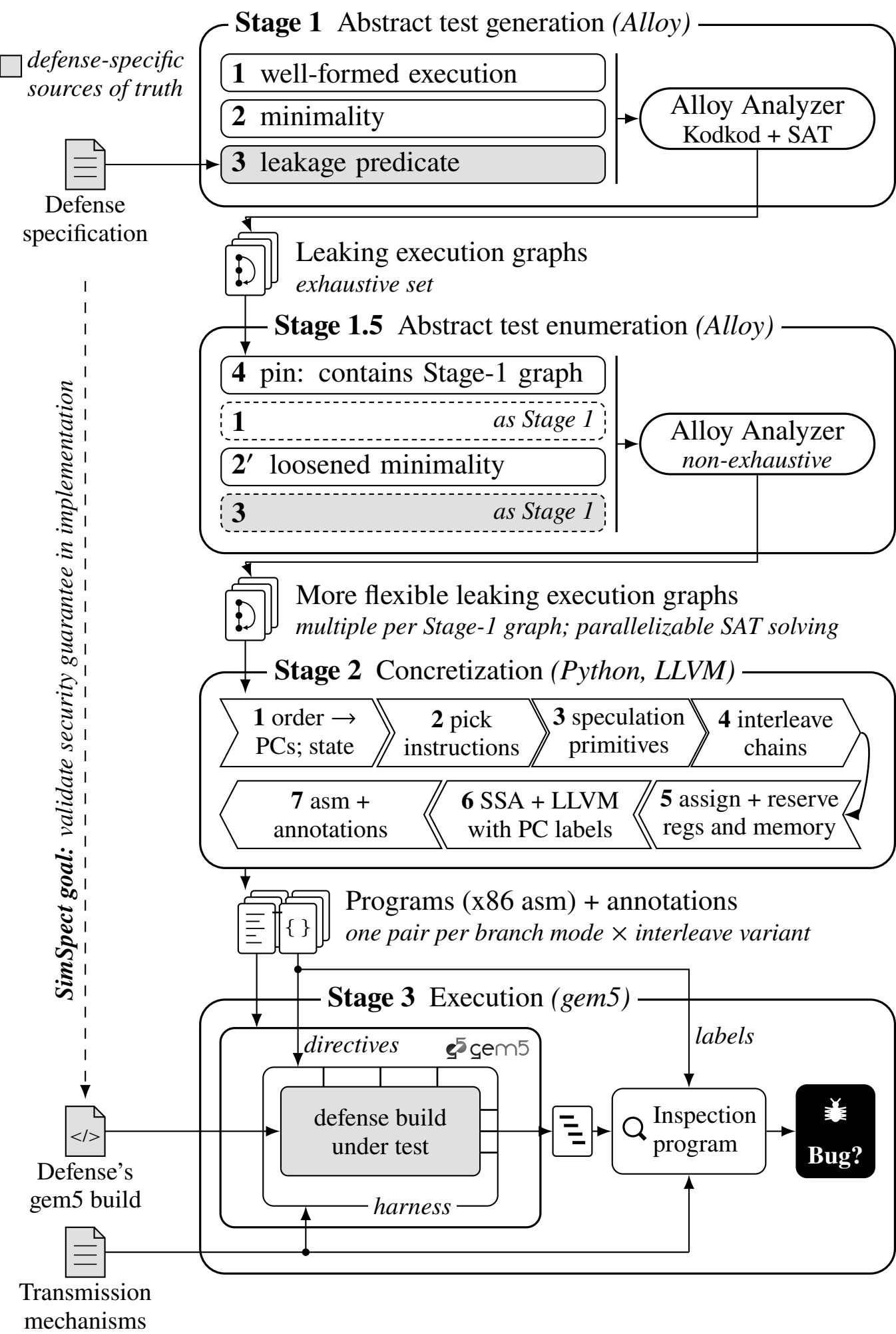

defense-specific sources of truth
Stage 1 Abstract test generation (Alloy)
1 well-formed execution
2 minimality
3 leakage predicate
Alloy Analyzer
Kodkod + SAT
Defense specification
Leaking execution graphs
exhaustive set
Stage 1.5 Abstract test enumeration (Alloy)
4 pin: contains Stage-1 graph
1 as Stage 1
2′ loosened minimality
3 as Stage 1
Alloy Analyzer
non-exhaustive
More flexible leaking execution graphs
multiple per Stage-1 graph; parallelizable SAT solving
Stage 2 Concretization (Python, LLVM)
1 order → PCs; state
2 pick instructions
3 speculation primitives
4 interleave chains
5 assign + reserve regs and memory
6 SSA + LLVM with PC labels
7 asm + annotations
Programs (x86 asm) + annotations
one pair per branch mode × interleave variant
Stage 3 Execution (gem5)
directives
gem5
defense build under test
harness
labels
Inspection program
Bug?
SimSpect goal: validate security guarantee in implementation
Defense's gem5 build
Transmission mechanisms